Class 9 Mathematics — Exercise 6.1

Basic Concept

(1) Sexagesimal System (English System) — Q1 to Q4

Use: Measurement of an angle.

(i) In sexagesimal system, an angle is measured in **Degrees, Minutes and Seconds**:

$$D^\circ \quad M' \quad S''$$

Degree Minute Second

A complete circle is divided into 360 equal pieces; one piece = 1°. So a complete rotation = 360° (quarter turns: $90^\circ, 180^\circ, 270^\circ, 360^\circ$).

(ii) Relation with each other:

$$1^\circ = 60' \quad \text{(one degree = 60 minutes)}$$

$$1' = 60'' \quad \text{(one minute = 60 seconds)}$$

$$\boxed{1^\circ = 60' = 3600''}$$

(iii) Conversion:

$$D^\circ \xrightarrow{\times 60} M' \xrightarrow{\times 60} S'' \qquad S'' \xrightarrow{\div 60} M' \xrightarrow{\div 60} D^\circ$$

(2) Radian Measure or Circular System — Q5 to Q7

$$2\pi \text{ radians} = 360^\circ$$

$$\pi \text{ radians} = 180^\circ$$

$$\boxed{1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ} \qquad \boxed{\frac{\pi}{180} \text{ radians} = 1^\circ}$$

$$\text{Degree} \xrightarrow{\times \frac{\pi}{180}} \text{Radian} \qquad \text{Radian} \xrightarrow{\times \frac{180}{\pi}} \text{Degree}$$

Radian:

- The ratio between length of circular arc and radius is called radian.
- If length of an arc is equal to radius of circle, then their ratio is **1 radian**.

Question 1

Convert the following measure of angles in seconds.

(i) 5° **(ii)** $30'$ **(iii)** $10^\circ 30'$ **(iv)** $20^\circ 20' 20''$

(i) 5°

$$= (5 \times 60 \times 60)''$$

$$\boxed{= 18000''}$$

(ii) $30'$

$$= (30 \times 60)''$$

$$\boxed{= 1800''}$$

(iii) $10^\circ 30'$

$$= 10^\circ + 30'$$

$$= (10 \times 60 \times 60)'' + (30 \times 60)''$$

$$= 36000'' + 1800''$$

$$\boxed{= 37800''}$$

(iv) $20^\circ 20' 20''$

$$= 20^\circ + 20' + 20''$$

$$= (20 \times 60 \times 60)'' + (20 \times 60)'' + 20''$$

$$= 72000'' + 1200'' + 20''$$

$$\boxed{= 73220''}$$

Question 2

Convert the following measure of angles in minutes.

(i) 75° **(ii)** $120''$ **(iii)** $50^\circ 40'$ **(iv)** $30^\circ 30' 30''$

(i) 75°

$$= (75 \times 60)'$$

$$\boxed{= 4500'}$$

(ii) $120''$

$$= \left(\frac{120}{60}\right)'$$

$$\boxed{= 2'}$$

(iii) $50°40'$

$$= 50° + 40'$$

$$= (50 \times 60)' + 40'$$

$$= 3000' + 40'$$

$$\boxed{= 3040'}$$

(iv) $30°30'30''$

$$= 30° + 30' + 30''$$

$$= (30 \times 60)' + 30' + \left(\frac{30}{60}\right)'$$

$$= 1800' + 30' + 0.5'$$

$$\boxed{= 1830.5'}$$

Question 3

Convert the following measure of angles in degrees and write the answer correct to 4 decimal places.

(i) $135'$ **(ii)** $150''$ **(iii)** $60°60'$ **(iv)** $45°45'45''$

(i) $135'$

$$= \left(\frac{135}{60}\right)°$$

$$\boxed{= 2.2500°}$$

(ii) $150''$

$$= \left(\frac{150}{60 \times 60}\right)° = \left(\frac{150}{3600}\right)°$$

$$= 0.04166°$$

$$\boxed{= 0.0417°}$$

(iii) $60^\circ 60'$

$$= 60^\circ + 60'$$

$$= 60^\circ + \left(\frac{60}{60}\right)^\circ$$

$$= 60^\circ + 1^\circ$$

$$\boxed{= 61.0000^\circ}$$

(iv) $45^\circ 45' 45''$

$$= 45^\circ + 45' + 45''$$

$$= 45^\circ + \left(\frac{45}{60}\right)^\circ + \left(\frac{45}{60 \times 60}\right)^\circ$$

$$= 45^\circ + 0.75^\circ + 0.0125^\circ$$

$$\boxed{= 45.7625^\circ}$$

Question 4

Write the following measures of angles in $D^\circ M' S''$.

(i) 60.125° **(ii)** $135.375''$ **(iii)** $60.85'$ **(iv)** 255.45°

(i) 60.125°

$$= 60^\circ + 0.125^\circ$$

$$= 60^\circ + (0.125 \times 60)'$$

$$= 60^\circ + 7.5'$$

$$= 60^\circ + 7' + 0.5'$$

$$= 60^\circ + 7' + (0.5 \times 60)''$$

$$= 60^\circ + 7' + 30''$$

$$\boxed{= 60^\circ\, 7'\, 30''}$$

(ii) $135.375''$

$$= \left(\frac{135.375}{60}\right)'$$

$$= 2.25625'$$

$$= 2' + 0.25625'$$

$$= 2' + (0.25625 \times 60)''$$

$$= 2' + 15.375''$$

$$= 2' + 15'' + 0.375''$$

$$= 2'\,15''$$

$$\boxed{= 0^\circ\,2'\,15''}$$

(iii) $60.85'$

$$= 60' + 0.85'$$

$$= 60' + (0.85 \times 60)''$$

$$= 60' + 51'' \qquad (\because 1^\circ = 60')$$

$$= 1^\circ + 51''$$

$$\boxed{= 1^\circ\,0'\,51''}$$

(iv) 255.45°

$$= 255^\circ + 0.45^\circ$$

$$= 255^\circ + (0.45 \times 60)'$$

$$= 255^\circ + 27'$$

$$\boxed{= 255^\circ\,27'\,0''}$$

Question 5

Convert the following in radians. Write the answer in terms of π.

(i) 45° **(ii)** 150° **(iii)** $60^\circ 30'$ **(iv)** 120°

$$\text{Degree} \xrightarrow{\times \frac{\pi}{180}} \text{Radian}$$

(i) 45°

$$= 45 \times \frac{\pi}{180} \text{ rad}$$

$$\boxed{= \frac{\pi}{4} \text{ rad}}$$

(ii) 150°

$$= 150 \times \frac{\pi}{180} \text{ rad}$$

$$\boxed{= \frac{5\pi}{6} \text{ rad}}$$

(iii) $60^\circ 30'$

$$= 60^\circ + 30'$$

$$= 60^\circ + \left(\frac{30}{60}\right)^\circ$$

$$= 60^\circ + 0.5^\circ = 60.5^\circ$$

$$= 60.5 \times \frac{\pi}{180}$$

$$\boxed{= \frac{121\pi}{360} \text{ rad}}$$

(iv) 120°

$$= 120 \times \frac{\pi}{180} \text{ rad}$$

$$\boxed{= \frac{2\pi}{3} \text{ rad}}$$

Question 6

Convert the following in radians. Use $\pi = 3.1416$.

(i) 270° **(ii)** 60° **(iii)** $180^\circ 45'$ **(iv)** $75^\circ 30' 45''$

$$\text{Degree} \xrightarrow{\times \frac{\pi}{180}} \text{Radian}$$

(i) 270°

$$= 270 \times \frac{\pi}{180} = \frac{3\pi}{2}$$

$$= \frac{3(3.1416)}{2} = 4.7124$$

$$\boxed{= 4.71 \text{ rad}}$$

(ii) 60°

$$= 60 \times \frac{\pi}{180} = \frac{\pi}{3}$$

$$= \frac{3.1416}{3} = 1.0472$$

$$\boxed{= 1.05 \text{ rad}}$$

(iii) $180^\circ 45'$

$$= 180^\circ + 45'$$

$$= 180^\circ + \left(\frac{45}{60}\right)^\circ$$

$$= 180^\circ + 0.75^\circ = 180.75^\circ$$

$$= 180.75 \times \frac{\pi}{180} = 180.75 \times \frac{3.1416}{180}$$

$$\boxed{= 3.15 \text{ rad}}$$

(iv) $75^\circ 30' 45''$

$$= 75^\circ + 30' + 45''$$

$$= 75^\circ + \left(\frac{30}{60}\right)^\circ + \left(\frac{45}{60 \times 60}\right)^\circ$$

$$= 75^\circ + 0.5^\circ + 0.0125^\circ = 75.5125^\circ$$

$$= 75.5125 \times \frac{\pi}{180} = 75.5125 \times \frac{3.1416}{180}$$

$$\boxed{= 1.32 \text{ rad}}$$

Question 7

Write the following radian measures of angle in $D^\circ M' S''$.

(i) $\frac{\pi}{4}$ **(ii)** $\frac{5\pi}{6}$ **(iii)** $\frac{\pi}{12}$ **(iv)** $\frac{7\pi}{40}$ **(v)** 1 **(vi)** 3.1416 **(vii)** 12π **(viii)** 5

$$\text{Radian} \xrightarrow{\times \frac{180}{\pi}} \text{Degree}$$

(i) $\frac{\pi}{4}$

$$= \frac{\pi}{4} \times \frac{180}{\pi}$$

$$\boxed{= 45^\circ}$$

(ii) $\frac{5\pi}{6}$

$$= \frac{5\pi}{6} \times \frac{180}{\pi}$$

$$\boxed{= 150^\circ}$$

(iii) $\frac{\pi}{12}$

$$= \frac{\pi}{12} \times \frac{180}{\pi}$$

$$\boxed{= 15^\circ}$$

(iv) $\frac{7\pi}{40}$

$$= \frac{7\pi}{40} \times \frac{180}{\pi} = \frac{63}{2}$$

$$= 31.5^\circ$$

$$= 31^\circ + 0.5^\circ$$

$$= 31^\circ + (0.5 \times 60)'$$

$$= 31^\circ + 30'$$

$$\boxed{= 31^\circ\, 30'}$$

(v) 1 radian

$$= 1 \times \frac{180}{\pi} = 1 \times \frac{180}{3.1416}$$

$$= 57.2956^\circ$$

$$= 57^\circ + 0.2956^\circ$$

$$= 57^\circ + (0.2956 \times 60)'$$

$$= 57^\circ + 17.736'$$

$$= 57^\circ + 17' + 0.736'$$

$$= 57^\circ + 17' + (0.736 \times 60)''$$

$$= 57^\circ + 17' + 44.16''$$

$$\boxed{\approx 57^\circ\, 17'\, 44''}$$

(vi) 3.1416 radians

$$= 3.1416 \times \frac{180}{\pi} = 3.1416 \times \frac{180}{3.1416}$$

$$\boxed{= 180^\circ}$$

(vii) 12π

$$= 12\pi \times \frac{180}{\pi}$$

$$\boxed{= 2160^\circ}$$

(viii) 5 radians

$$= 5 \times \frac{180}{\pi} = 5 \times \frac{180}{3.1416}$$

$$= 286.4782^\circ$$

$$= 286^\circ + 0.4782^\circ$$

$$= 286^\circ + (0.4782 \times 60)'$$

$$= 286^\circ + 28.692'$$

$$= 286^\circ + 28' + 0.692'$$

$$= 286^\circ + 28' + (0.692 \times 60)''$$

$$= 286^\circ + 28' + 41.52''$$

$$\boxed{\approx 286^\circ\, 28'\, 42''}$$